MODEL QUESTION PAPER : 2024-25

Course Code: BCSSS602

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VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi

Gokulam, 3rd Stage, Mysuru 570 002

Sixth Semester B.E. Examinations

COURSE NAME: SYSTEM SOFTWARE AND COMPILER DESIGN

Duration: 3 hours **Max. Marks: 100**

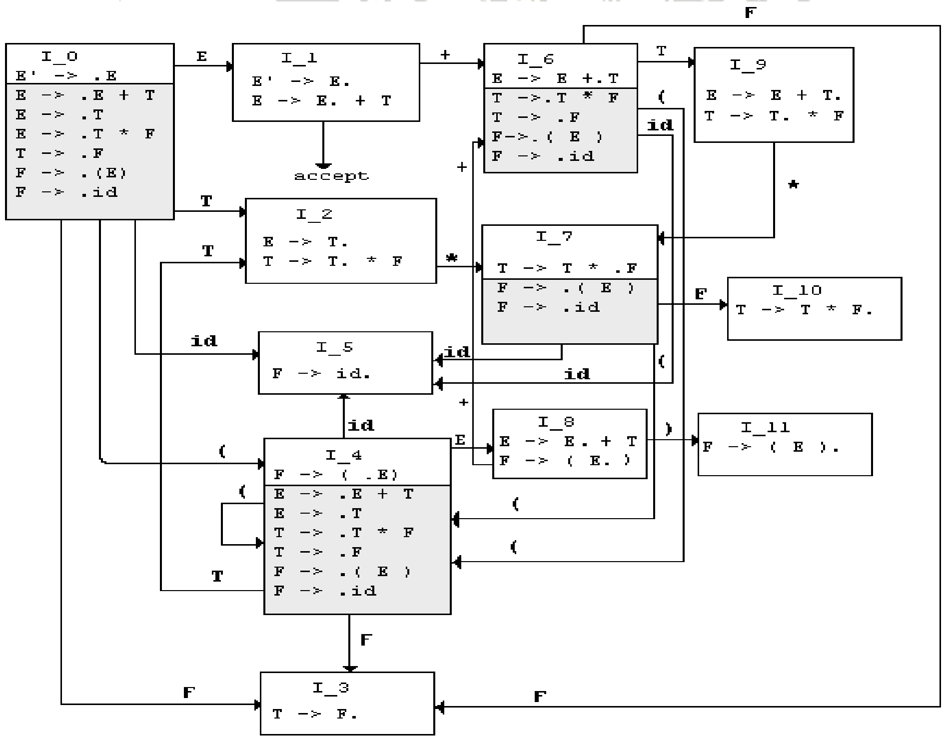
INSTRUCTION TO STUDENTS

1) Answer <u>One Full</u> question from each module.
2) Three module questions are compulsory and remaining two modules will have internal choice.
3) Each module carries 20 marks.

Q. No.	Module-I	Marks	BL	CO/PO
1.(a)	Explain the different phases of compiler and describe the translation of the statement “Position = initial + rate *6 ” at each phase.	10	L2	CO1/PO1
1.(b)	Construct the transition diagram for recognizing lexemes matching the token RELOP along with a sketch of implementation for the same.	10	L3	CO2/PO1
	Module-II			
2.(a)	Construct first and follow set for the given grammar. Also, generate the predictive parser table for the same. E → T E’ E’ → + T E’ | ϵ T → F T’ T’ → * F T’ | ϵ F → (E) | id	10	L3	CO2/PO1
2.(b)	Verify whether the given grammars are suitable for non-recursive predictive parsing. If not, state the reasons and make the grammars suitable for top-down parsing. i) S → OS1 | O1 ii) S → SS+ | SS* | a	10	L4	CO3/PO2
	(OR)			
3.(a)	(see parse table below) Using the above parse table, parse the following strings i) num + num ii) num + (num+num)	10	L3	CO2/PO1

Terminal/Input	num	(	+	)	$
S	S → ES^l	S→ ES^l			
S^l			S^l → +S	S^l → ε	S^l → ε
E	E → num	E → (S)			

Q. No.		Marks	BL	CO/PO
3.(b)	Verify whether any conflicts arise during parsing the string (a,(a,a)) in SR parser for the below grammar.	10	L4	CO3/PO2

<table>
<tr><td></td><td>S → (L) | a
L → L,S | S</td><td></td><td></td><td></td></tr>
<tr><td colspan="5">Module-III</td></tr>
<tr><td>4.(a)</td><td>Explain three-address code with an example.</td><td>5</td><td>L2</td><td>CO1/PO1</td></tr>
<tr><td>4.(b)</td><td>Construct LR(0) automaton for the given grammar
E → E+T | T
T → T * F | F
F → id</td><td>10</td><td>L3</td><td>CO2/PO1</td></tr>
<tr><td>4.(c)</td><td>For the given syntax directed definition (SDD) as shown in Table 1, verify the dependency between the variables in the expression 3 * 5 + 4 using S-attributed SDT and construct topological ordering.
Table 1. Syntax Directed Definition
<table>
<tr><th></th><th>PRODUCTION</th><th>SEMANTIC RULES</th></tr>
<tr><td>1)</td><td>$L \to E\ \mathbf{n}$</td><td>$L.val = E.val$</td></tr>
<tr><td>2)</td><td>$E \to E_1 + T$</td><td>$E.val = E_1.val + T.val$</td></tr>
<tr><td>3)</td><td>$E \to T$</td><td>$E.val = T.val$</td></tr>
<tr><td>4)</td><td>$T \to T_1 * F$</td><td>$T.val = T_1.val \times F.val$</td></tr>
<tr><td>5)</td><td>$T \to F$</td><td>$T.val = F.val$</td></tr>
<tr><td>6)</td><td>$F \to (E)$</td><td>$F.val = E.val$</td></tr>
<tr><td>7)</td><td>$F \to \mathbf{digit}$</td><td>$F.val = \mathbf{digit}.lexval$</td></tr>
</table></td><td>5</td><td>L4</td><td>CO3/PO2</td></tr>
<tr><td colspan="5">(OR)</td></tr>
<tr><td>5.(a)</td><td>Explain syntax directed definition (SDD) with an example.</td><td>5</td><td>L2</td><td>CO1/PO1</td></tr>
<tr><td>5.(b)</td><td>Construct goto and action table for the given LR (0) automaton as shown in Figure 1. Also, parse the string id * id + id.

</td><td>10</td><td>L3</td><td>CO2/PO1</td></tr>
</table>

<table>
<tr><td></td><td>Figure 1. LR(0) automaton</td><td></td><td></td><td></td></tr>
<tr><td>5.(c)</td><td>Verify the given expression using Direct Acyclic Graph (DAG) and construct three address code.
a = b * - c + b * - c</td><td>5</td><td>L4</td><td>CO3/PO2</td></tr>
<tr><td colspan="5">Module-IV</td></tr>
<tr><td>6.(a)</td><td>Explain the issues in the design of a code generator.</td><td>5</td><td>L2</td><td>CO1/PO1</td></tr>
<tr><td>6.(b)</td><td>Describe the machine architecture features of SIC assembler.</td><td>5</td><td>L2</td><td>CO1/PO1</td></tr>
<tr><td>6.(c)</td><td>Generate assembly code for the below three address code using three registers R1, R2 & R3
t = a – b
u = a – c
v = t + u
a = d
d = v + u</td><td>10</td><td>L3</td><td>CO2/PO1</td></tr>
<tr><td colspan="5">Module- V</td></tr>
<tr><td>7.(a)</td><td>Explain program relocation using Bit Mask.</td><td>5</td><td>L2</td><td>CO1/PO1</td></tr>
<tr><td>7.(b)</td><td>Generate the object program for the following SIC/XE program.
Given CLEAR= B4, LDA=00, LDB=68, ADD=18, TIX=2C, JLT=38, STA=0C
<table>
<tr><td>Label</td><td>Opcode</td><td>Operand</td></tr>
<tr><td>SUM</td><td>START</td><td>0</td></tr>
<tr><td>FIRST</td><td>CLEAR</td><td>X</td></tr>
<tr><td></td><td>LDA</td><td>#0</td></tr>
<tr><td></td><td>+LDB</td><td>#TOTAL</td></tr>
<tr><td></td><td>BASE</td><td>TOTAL</td></tr>
<tr><td>LOOP</td><td>ADD</td><td>TABLE, X</td></tr>
<tr><td></td><td>TIX</td><td>COUNT</td></tr>
<tr><td></td><td>JLT</td><td>LOOP</td></tr>
<tr><td></td><td>STA</td><td>TOTAL</td></tr>
<tr><td>COUNT</td><td>RESW</td><td>1</td></tr>
<tr><td>TABLE</td><td>RESW</td><td>2000</td></tr>
<tr><td>TOTAL</td><td>RESW</td><td>1</td></tr>
<tr><td></td><td>END</td><td>FIRST</td></tr>
</table></td><td>15</td><td>L3</td><td>CO2/PO1</td></tr>
</table>